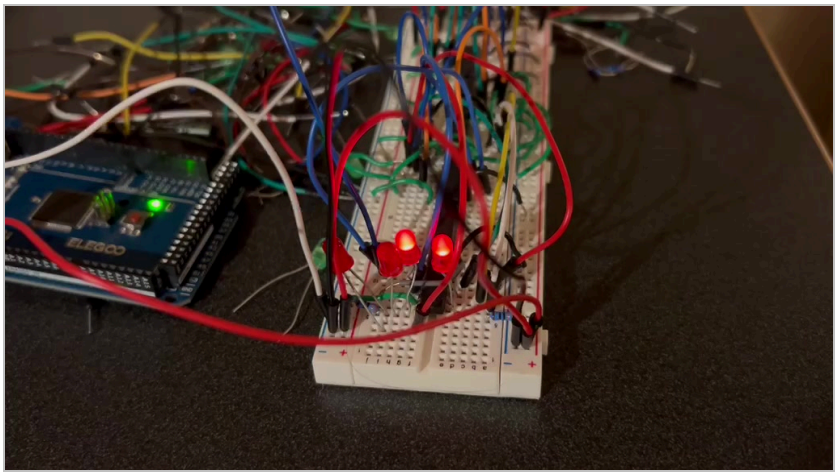
4-bit LED counter from logic gates

Four LEDs, one per bit, driven by TI 74HC-series AND, OR and NOT gates and a 555 timer clock on a breadboard to show a set sequence of numbers on repeat.

Key parameters

Parameter	Value
Logic	AND / OR / NOT 74HC
Clock	555 timer

Data status: Built on a breadboard and working. No measurements are recorded on this page.



Breadboard with four red LEDs showing a 4-bit number.

Online version: https://saosith.github.io/projects/led-4bit-counter/

1 What it does

Each of the four LEDs is one bit of a binary number. I wired TI 74HC-series logic chips, such as the SN74HC08 (AND), SN74HC32 (OR) and SN74HC04 (NOT), D flip-flops such as the SN74HC74 to hold the current number, and a 555 timer as the clock, so the LEDs step through a set sequence of numbers and then repeat. The blue Elegoo board in the photo only supplies power from my laptop over USB; the 555 sets the timing.

2 How it works

The 555 timer runs in astable mode, which means it has no stable state and keeps switching on its own. A capacitor charges and discharges through resistors, and each time it crosses a threshold the 555 flips its output between high and low. The result is a steady on/off pulse signal, and each pulse is one clock tick.

Four D flip-flops, one per LED, hold the current 4-bit number. The gates read that number and work out the next number in the sequence. On each clock tick, the flip-flops load the next number, the LEDs change, and the gates start working out the one after it.

Status: built and working on a breadboard. The number sequence will be added here.